SEMANTIC AND INSTANCE-AWARE PIXEL-ADAPTIVE CONVOLUTION FOR PANOPTIC SEGMENTATION

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ABSTRACT

Although the weight-sharing property of convolution is one of the major reasons for the success of convolution neural networks, the content-agnostic operation is insufficient for several tasks requiring content-adaptive processing, including panoptic segmentation. Inspired by several recent works on content-adaptive convolutions, we introduce the Guided-PAKA, the first content-adaptive convolution method specialized for panoptic segmentation. Specifically, GuidedPAKA learns the pixel-adaptive kernel attention consisting of the channel and spatial kernel attentions. Instead of commonly used self-attention operation, we guide the channel and spatial kernel attentions using their respective supervision signals, *i.e.*, semantic segmentation maps and local instance affinities. Consequently, these kernel attentions extract features helpful for panoptic segmentation. Experimental results show that the proposed GuidedPAKA improves the performance of panoptic segmentation when integrated into the baseline model.

Index Terms— Instance segmentation, panoptic segmentation, pixel-adaptive convolution, semantic segmentation

1. INTRODUCTION

Standard convolution that constitutes a convolutional neural network (CNN) calculates a feature for every pixel by applying shared weights to its receptive field. Although such content-agnostic processing contributes to parameter reduction and translation equivalence, it is insufficient for tasks requiring a holistic understanding of images, such as image segmentation. Therefore, various research has been conducted to extract global contextual information by adjusting features or weights. Several early studies presented network modules that utilize multiple sampling rates or semantic region representation for feature re-weighting [1–4]. Meanwhile, some recent studies attempted to re-weight shared weights to generate content-adaptive features [5–8]. For example, Hang *et al.* [5] suggested pixel adaptive convolution (PAC), which adapts standard convolution with the spatially varying kernel obtained from the guidance image. Since PAC changes the shared weights to pixel-adaptive weights using the contextual kernel, it helps to make discriminating features for semantic segmentation. Furthermore, Sagong *et al.* [6] introduced the pixel-adaptive kernel attention (PAKA) that applies pixel-adaptive kernels using directional and channel attention modules. Specifically, the directional attention module captures directional features from different kernel offsets, while the channel attention module aggregates inter-channel relationships. With PAKA, the pixel-adaptive convolution predicts directions that provide pertinent information to propagate semantic-aware features.

In this paper, we propose a novel pixel-adaptive convolution method called GuidedPAKA for panoptic segmentation. Since panoptic segmentation is a task of assigning semantic class and instance-id to every pixel in an image, convolution operations need to be both semantic-adaptive and instance-adaptive. GuidedPAKA thus learns semantic and instance-aware pixel-adaptive kernels using their dedicated supervisions, *i.e.*, semantic segmentation labels and instance affinity labels, respectively. Experimental results demonstrate the superiority of the proposed GuidedPAKA over its baseline PAKA and other adaptive convolution modules when applied to panoptic segmentation.

2. PROPOSED METHOD

Fig. 1(a) illustrates the structure of the proposed Guided-PAKA. GuidedPAKA derives a guided kernel from the channel and spatial attention units. Different from the standard PAKA, which learns kernel attention without supervision, the channel and spatial attention units are supervised by semantic segmentation labels and instance affinity labels, respectively. The guided kernel is then element-wisely multiplied with the shared weights to generate pixel-adaptive convolutional kernels. Although GuidedPAKA can be applied to any CNN structures, we embed GuidedPAKA to the feature pyramid network (FPN) [9], which is one of the most commonly used pixel decoders in the segmentation task. Fig. 1(b) shows the structure of the GuidedPAKA-embedded FPN. More details will be described in the following subsections.

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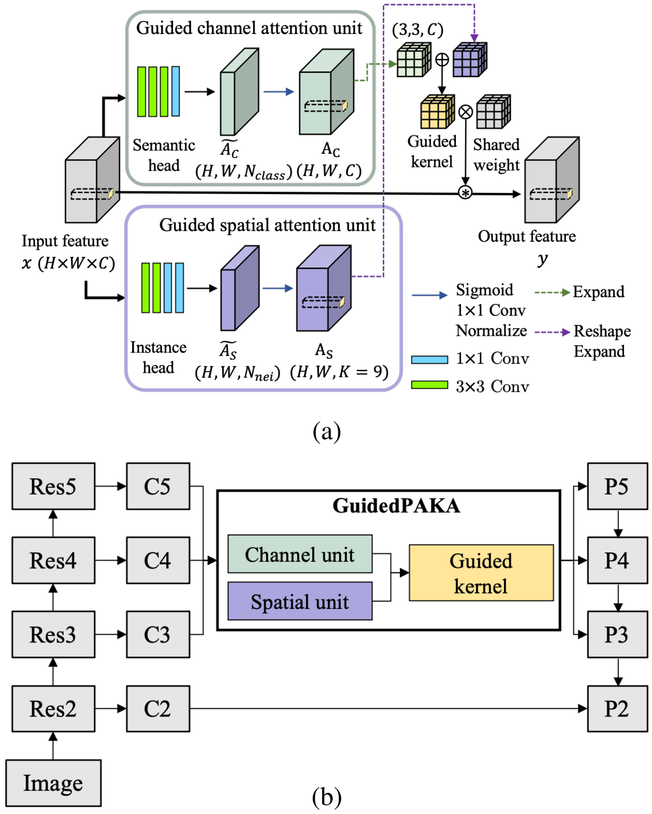

Fig. 1. (a) The structure of GuidedPAKA and (b) GuidedPAKA-embedded FPN. GuidedPAKA is shared to extract Ps and generates guided kernel attention for every pixels.

2.1. Pixel-Adaptive Kernel Attention

The standard convolution for each pixel coordinate p in the output feature map $y \in R^{H \times W}$ using the input feature map $x \in R^{H \times W \times C}$ can be written as

$$y(p) = \sum_{j=1}^{C} \sum_{k=1}^{K} x(p + p_k, j) \cdot w(k, j), \tag{1}$$

where $w(k,\ j)$ denotes the shared weight for the k-th spatial location and j-th channel. H and W are the height and width of the feature map, respectively, K and C represent the size of the kernel and the number of channels, respectively, and p_k is the pre-specified offset for the k-th location. For example, for the 3×3 convolution, $K = 9$ and $p_k \in \{(-1, -1), (-1, 0), ..., (1, 1)\}$. Note that $w(k, j)$ is not a function of p, *i.e.*, the same weight is applied to every pixel coordinate regardless of the image content.

Sagong *et al.* proposed PAKA [6], which multiplies a pixel-adaptive learnable kernel attention A to the shared weight w as follows:

$$y(p) = \sum_{j=1}^{C} \sum_{k=1}^{K} x(p + p_k, j) \cdot w(k, j) \cdot A(p, k, j), \tag{2}$$

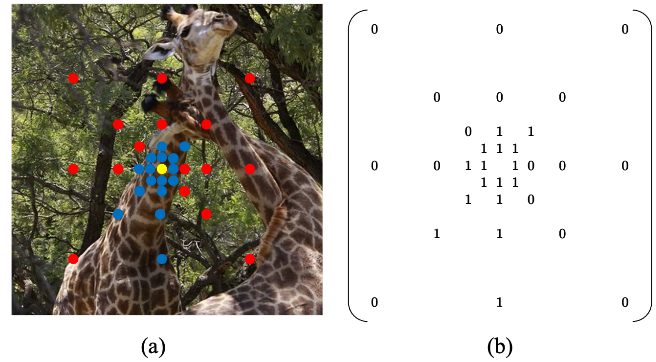

Fig. 2. Illustration for local instance affinities [3]: (a) Locations of the neighbors with different dilations, where the blue and red pixels indicate pixels belonging to the same and different instance to the center (yellow) pixel, respectively, and (b) the corresponding binary labels used as the ground-truth for $\tilde{A}_S$.

where $A(p, k, j)$ indicates kernel attention at the position $p + p_k$ of the j-th channel. A is constructed as an integration of the channel attention $A_C \in R^{H \times W \times C}$ and spatial attention $A_S \in R^{H \times W \times K}$. In particular, A_C and A_S are added with the tensor broadcast to match their dimension, followed by the tanh activation function to control the distribution of the attention. As a result, the kernel attention can be expressed as

$$A(p, k, j) = 1 + \tanh(A_S(p, k) + A_C(p, j)), \tag{3}$$

where $A_S(p, k)$ and $A_C(p, j)$ represent the spatial kernel attention at the offset p_k and the channel attention at the j-th channel of the pixel location p, respectively.

The PAKA [6] learns the kernel attention without supervision. Unlike PAKA, GuidedPAKA learns attention from their dedicated supervision signals, *i.e.*, semantic and instance affinity labels, respectively. Although the same adaptive convolution operation in (2) is used for GuidedPAKA, its explicit guidance for the kernel attention leads to panoptic segmentation-aware pixel-adaptive kernels.

2.2. Guided Channel Attention

Different from recent works [6, 10] that derive channel attention from inter-channel dependencies without supervision, the proposed GuidedPAKA obtains channel attention with the guidance of semantic segmentation labels. As shown in Fig. 1(a), the semantic head conducts semantic segmentation, leading to semantic-aware pixel-adaptive convolution when combined with the shared weights.

Let x be the input feature of the guided channel attention unit. The input feature is processed by the semantic head consisting of three 3×3 convolution layers and one 1×1 convolution layer, resulting in the feature $\tilde{A}_C \in R^{H \times W \times N_{class}}$, where N_{class} is the number of semantic classes. We supervise the class prediction $\tilde{A}_C$ by the ground-truth semantic segmentation labels with the focal loss [11]. Consequently, the pixels belonging to the same (different) semantic class can have similar (dissimilar) channel attention weights. A single 1×1 convolution layer is applied to $\tilde{A}_C$, resulting in A_C with the channel dimension of C. And then, SyncBatchNorm [4] is used to keep the kernel distribution.

2.3. Guided Spatial Attention

Inspired by [3], we define a set of neighboring pixels of p as

$$N(p) = \bigcup_{d \in D} N_d(p), \tag{4}$$

where $N_d(p)$ is the set of eight neighbors of p with dilation d. D is the set of dilations, chosen as $D = \{1, 2, 4, 8\}$ as illustrated in Fig. 2(a). The guided spatial attention unit predicts instance affinities of the neighboring pixels, resulting in instance-aware pixel-adaptive convolution. Like the guided channel attention unit, the input feature x is processed by instance head to predict whether neighboring pixels $\tilde{A}_S \in R^{H \times W \times N_{nei}}$ belong to the same instance, where N_{nei} is the number of elements in $N(p)$. The instance head is composed of two 3×3 convolution layers and two 1×1 convolution layers. We supervise feature $\tilde{A}_S$ using the ground-truth instance affinity labels. As illustrated in Fig. 2(b), the ground-truth instance affinity is given as 32 (= 8 neighbors $\times$ 4 dilations)-dimensional binary vector for each pixel, and the focal loss is used for the training. As a result, the neighboring pixels belonging to the same (different) instance can receive high (low) attention scores. A single 1×1 convolution layer is applied to output A_S with K channels, followed by SyncBatchNorm.

2.4. GuidedPAKA-embedded Feature Extractor

Panoptic segmentation networks generally consist of a backbone for feature extraction and a mask prediction module [12–14]. Most research has focused on designing an effective mask prediction module, but a few representative backbone architectures, especially the FPN [9], have been dominantly used in the literature [12, 13, 15]. In order to demonstrate broad applicability, we insert GuidedPAKA into the FPN structure, as shown in Fig. 1(b).

FPN consists of a ResNet-based [16] pixel encoder and a pixel decoder with lateral connections. Let $\{C_2, C_3, C_4, C_5\}$ denote the features with the strides of $\{4, 8, 16, 32\}$ pixels with respect to the input image, respectively. In the standard FPN, these features are passed through a 3×3 convolution layer respectively, resulting in the features $\{P_2, P_3, P_4, P_5\}$. We replace the three 3×3 convolutional layers that extract $\{P_3, P_4, P_5\}$ with GuidedPAKA to utilize semantic and neighbor information from diverse feature scales. In addition, the parameters of the guided channel attention and spatial attention units are shared across scales to minimize the memory overhead caused by GuidedPAKA.

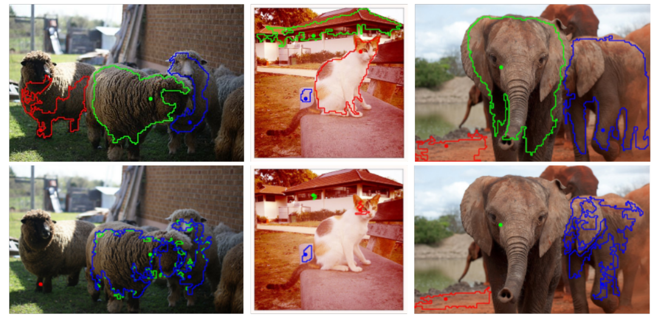

Fig. 3. Clustering results of the neighboring pixels with high attention scores for the marked pixels for GuidedPAKA (top) and PAKA (bottom).

2.5. Loss for GuidedPAKA

In each g-th GuidedPAKA, we resize the ground-truth to the same resolution as the input feature. The resized ground-truth semantic segmentation and local affinity prediction labels are denoted as Y^g_{sem} and Y^g_{nei}, respectively. The loss function for GuidedPAKA, denoted as L_{gp}, is expressed as

$$L_{gp} = \sum_{g=1}^{3} \lambda_{sem} F(\tilde{A}^g_C, Y^g_{seg}) + \lambda_{nei} F(\tilde{A}^g_S, Y^g_{nei}), \tag{5}$$

where F represents the focal loss [11]. λ_{sem} and λ_{nei} are the hyper-parameters for the two loss terms. As a result, L_{gp} is combined with the original loss functions of the panoptic segmentation models embedded with GuidedPAKA.

2.6. Analysis of GuidedPAKA

To understand the effects of the proposed GuidedPAKA, we visualize the kernel attention area of A for the marked pixels, as shown in Fig. 3. Specifically, let p be a marked pixel. The attention score map a is then defined as $a(p) = \max(A(p, :, :))$. We then cluster neighboring pixels of p using $a(p)$. Specifically, we initialize the set with p. Then, among eight neighbors of p, we add neighboring pixels to the set if their attention values in a are higher than the threshold, empirically chosen as 0.8. The set grows repeated by following the same procedure until convergence. In this manner, we can cluster neighboring pixels that have high attention values to the marked pixels. As shown in Fig. 3, GuidedPAKA can assign similar attention scores to the pixels on the same instance, whereas PAKA fails to find directions to propagate semantic-aware features.

Table 1. Performance comparison of convolution modules on the COCO validation set using the PanopticFCN as a baseline network. † indicates the model extended to match the number of parameters to GuidedPAKA. ⋆ represents the model without sharing parameters in the channel and spatial units for different feature scales.

Method	PQ	PQ^{th}	PQ^{st}	AP	mIOU	Param
PanopticFCN	44.4	50.3	35.5	36.0	44.1	38.6M
+ DynamicConv [17]	44.6	50.5	35.8	36.3	44.1	51.2M
+ DeformConv [18]	44.4	50.2	35.6	36.2	43.6	39.0M
+ PAC [5]	44.5	50.5	35.5	36.4	44.2	38.8M
+ PAKA [6]	44.3	50.2	35.5	36.2	43.9	39.9M
+ PAKA†	44.7	50.9	35.4	36.6	43.7	42.1M
+ channel unit	44.9	50.7	36.2	36.9	44.4	40.7M
+ spatial unit	44.7	50.7	35.5	36.4	43.9	40.9M
+ GuidedPAKA	**45.3**	**51.0**	36.9	**36.8**	44.7	41.9M
+ GuidedPAKA⋆	**45.3**	50.7	**37.0**	**36.8**	**45.2**	48.1M

Table 2. Performance comparison using Mask2Former as a baseline network.

Dataset	Pixel decoder	PQ	AP	mIOU	Param
COCO val	FPN [9]	50.7	**41.5**	46.6	50.0M
	+ GuidedPAKA	**52.3**	41.4	**63.7**	52.9M
ADE20K val	FPN [9]	36.8	23.0	44.2	50.0M
	+ GuidedPAKA	**39.1**	**23.8**	**44.9**	52.8M

3. EXPERIMENTAL RESULTS

To show the effectiveness of GuidedPAKA, we embedded it on PanopticFCN [12] and Mask2Former [15] as baseline networks with the ResNet50 pixel encoder. All experiments were conducted using Detectron2 [19] with PanopticFCN-Star-R50-3x.yaml and Maskformer2-R50-bs16-50ep.yaml configurations. Hyper-parameters λ_{sem} and λ_{nei} are set to 1. In Table 1, we compared GuidedPAKA with the other pixel-adaptive convolution methods on PanopticFCN for the COCO validation set, including DynamicConv [17], PAC [5], PAKA [6], and DeformableConv [18], by replacing the standard convolutions with each proposed convolution. GuidedPAKA shows superior performance, *i.e.*, 0.9% PQ, 0.8% AP, and 0.6% mIOU improvements over the baseline, respectively, with 3.3M parameter increases. In particular, the performance improvement over PAKA† demonstrates that supervised learning of kernel attention by using the dedicated labels helps generate content-aware kernels. We also conducted experiments without sharing the parameters of the two attention units across feature scales, which is indicated ⋆. GuidedPAKA⋆ shows a similar performance to GuidedPAKA but with 6.2M more parameters. It demonstrates that GuidedPAKA improves performance significantly with fewer parameters. Fig. 4 shows some results on the COCO validation set for panoptical segmentation performance comparison.

(a)
(b)
(c)
(d)
(e)

Fig. 4. Comparison of the panoptic segmentation results on the COCO validation set with PanopticFCN: (a) Input image, (b) ground-truth, and the results obtained using (c) standard convolution, (d) PAKA, and (e) GuidedPAKA.

Table 2 provides resulks using Mask2Former [15] as a baseline network. On the COCO and ADE20K datasets, FPN with GuidedPAKA exhibits higher performance than FPN. More results and code are available at the project webpage[1].

4. CONCLUSION

We proposed the first pixel-adaptive convolution for panoptic segmentation, named GuidedPAKA. GuidedPAKA learns the channel attention using the semantic segmentation labels, making pixels in the same semantic class be processed by similar convolution kernels. Moreover, GuidedPAKA learns the spatial attention using the local instance affinity labels indicating whether the center pixel and surrounding pixels belong to the same instance, leading to instance-aware convolution kernels. By integrating the channel and spatial attentions as pixel-adaptive kernel attention, we demonstrated that semantic-aware and instance-aware convolution kernels can be obtained, which are essential for effective panoptic segmentation. Using the two representative networks, PanopticFCN and Maskformer2, we showed that the performance of the existing models could be further improved by changing the standard convolution to the proposed GuidedPAKA.

[1]https://github.com/SUMIN97/GuidedPAKA

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